

Advanced (Habanero)

- Q1.** Solve $|x| = 5$
(A) $x = 5$
(B) $x = -5$
(C) $x = 5$ or $x = -5$
(D) $x = 0$
(E) $x = 10$
- Q2.** Solve $|2x| = 8$
(A) $x = 2$ or $x = -2$
(B) $x = 4$ or $x = -4$
(C) $x = 1$ or $x = -1$
(D) $x = 8$ or $x = -8$
(E) $x = 0$
- Q3.** Solve $|x - 3| = 2$
(A) $x = 1$ or $x = 5$
(B) $x = 3$ or $x = -3$
(C) $x = 2$ or $x = 4$
(D) $x = 0$ or $x = 6$
(E) $x = -1$ or $x = 7$
- Q4.** Solve $|x + 2| = 3$
(A) $x = -5$ or $x = 1$
(B) $x = -2$ or $x = 4$
(C) $x = 0$ or $x = 6$
(D) $x = -3$ or $x = 5$
(E) $x = -1$ or $x = 7$
- Q5.** Solve $|3x - 2| = 10$
(A) $x = 4$ or $x = -2$
(B) $x = 2$ or $x = 0$
(C) $x = 1$ or $x = -3$
(D) $x = 6$ or $x = -4$
(E) $x = 8$ or $x = -6$
- Q6.** Solve $|2x + 5| = 11$
(A) $x = 2$ or $x = -8$
(B) $x = 1$ or $x = -9$
(C) $x = 3$ or $x = -7$
(D) $x = 0$ or $x = -10$
(E) $x = 4$ or $x = -6$
- Q7.** Solve $|x - 2| + |x + 2| = 8$
(A) $x = 0$ or $x = 4$
(B) $x = -2$ or $x = 2$
(C) $x = -4$ or $x = 4$
(D) $x = 1$ or $x = 3$
(E) $x = 2$ or $x = -2$
- Q8.** Solve $|3x| - 2|2x| = -2$
(A) $x = 0$
(B) $x = 1$ or $x = -1$
(C) $x =$
 $\frac{23}{12}$ or $x = -\frac{23}{12}$
(C) $x =$
 $\frac{12}{13}$ or $x = -\frac{12}{13}$
(C) $x =$
 $\frac{13}{13}$ or $x = -\frac{13}{13}$

Open Ended Questions (FRQ)

Q9. Solve $|x^2 - 4| = 3$ and check for extraneous solutions

Q10. Solve $|2x + 1| = |x - 3|$ and check for extraneous solutions

Chef's Tips

- Tip 1: Emphasize checking for extraneous solutions
- Tip 2: Encourage students to sketch graphs to visualize absolute value functions
- Tip 3: Remind students to consider both positive and negative cases for absolute value equations